Project Proposal: Digital Twin-Driven DRL for Energy-Aware CNC Milling

1. Problem Statement

In modern CNC machining, selecting optimal cutting parameters (spindle speed, feed rate, and depth of cut) is critical for balancing production throughput with surface quality and tool longevity. Traditional meta-heuristic algorithms optimize these parameters statically, failing to account for the dynamic degradation of the cutting tool (wear and stochastic breakage) during multi-pass operations. Furthermore, modern manufacturing requires energy-efficient operations to reduce the environmental footprint.

This project proposes a Deep Reinforcement Learning (DRL) approach that formulates cutting parameter optimization as a Markov Decision Process (MDP). By interacting with a physics-based Digital Twin, the DRL agent learns a dynamic control policy to maximize Material Removal Rate (MRR), minimize energy consumption, guarantee surface roughness requirements, and prevent catastrophic tool breakage.

2. Detailed Digital Twin Environment

The environment acts as a physics-based data generator simulating multi-pass milling operations.

2.1 State Space (Observations)

The agent observes an 8-dimensional normalized state vector before each cutting pass. The state is represented as $S_t \in [0, 1]^8$:

Index	Feature	Notation	Unit	Description
0	Tool wear value	VB_i	mm	Current flank wear of the tool.
1	Remaining Allowance	κ	mm	Depth of material yet to be removed.
2	Previous spindle speed	n_{i-1}	RPM	Spindle speed used in the previous pass.
3	Previous feed rate	f_{i-1}	mm/r	Feed rate used in the previous pass.
4	Previous depth of cut	ap_{i-1}	mm	Depth used in the previous pass.
5	Previous MRR	MRR_{i-1}	mm^3/min	Material removal rate of the last pass.
6	Previous surface roughness	Ra_{i-1}	μm	Roughness achieved in the last pass.
7	Tool Breakage State	B_i	Boolean	1 if tool broke, 0 otherwise.

2.2 Action Space

The agent outputs a continuous 3-dimensional action vector mapped to the physical machine bounds:

Action	Parameter	Bounds	Unit
a_0	Spindle speed (n)	[2000, 8000]	RPM
a_1	Feed rate (f_r)	[0.05, 0.35]	mm/r
a_2	Depth of cut (a_p)	[0.1, 2.5]	mm

2.3 Reward Function

The reward function R_t guides the agent using a combination of dense immediate rewards and sparse terminal Pareto-based rewards:

- **Immediate Penalties:** -50 if $Ra > 1.6 \mu m$; -50 if $VB > 0.3 \text{ mm}$; -100 for catastrophic breakage.
- **Throughput (MRR):** $+ 0.5 \times (MRR / MRR_{max})$ to encourage faster cutting.
- **Energy Penalty:** $-w_2 \times (E_i / E_{max})$ (Novel addition to penalize excessive power draw).
- **Hypervolume (HV) Reward:** Calculated upon successful completion of the final finishing pass: $HV = (T_{term} / T_{ref}) \times (Ra_{term} / Ra_{ref}) \times 100.0$.

3. Reinforcement Learning Algorithm (PPO)

The chosen algorithm is Proximal Policy Optimization (PPO), implemented with a 2-layer MLP (64-64 hidden units) for both the Actor and Critic networks. PPO is an on-policy algorithm that strikes an excellent balance between sample efficiency and ease of tuning. Given the continuous nature of the action space and the complexity of the multi-objective delayed Hypervolume rewards, PPO provides the stable gradient updates necessary to learn a robust and safe machining policy without catastrophic forgetting.

4. Training Method

Training is conducted purely within the Digital Twin environment to allow for safe, rapid trial-and-error learning without risking physical machine damage.

1. **Vectorization:** Multiple instances (e.g., 4 to 8 parallel environments) are run to collect diverse experiences and decorrelate the training batches.
2. **Exploration vs Exploitation:** The PPO agent begins by taking random actions (within the continuous bounds) to explore the consequences of aggressive cutting on tool wear.
3. **Value Estimation & Policy Update:** The Critic network learns to estimate the Generalized Advantage Estimation (GAE) of current machining states, while the Actor network adjusts the Gaussian probability distribution of continuous actions to maximize expected returns while staying within a clipping threshold ($\epsilon = 0.2$).
4. **Convergence:** The agent is trained for 500,000 timesteps (equivalent to machining approximately 500,000 synthetic workpieces), ensuring the policy stabilizes and accurately adapts to varying degradation rates.

5. Inference Method (Evaluation)

During inference, the trained policy acts as a lightweight, real-time edge controller. Before every single cutting pass, the agent ingests the 8-dimensional state vector and outputs

optimal machining parameters in milliseconds.

To ensure the policy has generalized, inference is conducted via a **Massive Monte Carlo Random Sweep**. The agent is evaluated over hundreds of episodes where underlying environmental constants are highly randomized:

- Total machining allowance (κ) varies [2.0, 15.0] mm.
- Workpiece diameter (D) varies [40.0, 150.0] mm.
- Tool wear coefficients (C_{wear}) vary to simulate materials ranging from soft aluminum to hard titanium.

Appendix: Physics Formulations & Literature Borrowing

The core physics equations driving our Digital Twin environment are meticulously grounded in recent academic literature. We have adopted the core environment formulations from two primary papers while introducing a novel optimization objective.

What we borrowed from Mo et al.

(Reference: "DRL-based optimization for machining process: Cutting parameters dynamic optimization considering tool condition degradation" - J. Manuf. Processes, 2025)

- **Cutting Speed (v_c):** $v_c = (\pi \times D \times n) / 1000$ *(Eq. 5)*
- **Feed per Minute (f):** $f = n \times f_r$ *(Eq. 6)*
- **Cutting Time (t_{cut}):** $t_{cut} = L_{cut} / f$ *(Eq. 7)*
- **Material Removal Rate (MRR):** $MRR = v_c \times a_p \times f_r$ *(Eq. 27)*
- **Tool Wear Degradation (ΔVB):** The Taylor-extended wear model ($\Delta VB = C \cdot v_c^\alpha \cdot f_r^\beta \cdot a_p^\gamma \cdot t_{cut}$) is adapted directly from their tool wear constraint analysis *(Eq. 17)*.
- **Stochastic Breakage Logic:** The concept of assigning a probability of catastrophic failure when the tool is highly worn and pushed at a high MRR.

What we borrowed from Wang et al.

(Reference: "Machining parameter optimization for a batch milling system using multi-task deep reinforcement learning" - J. Manuf. Systems, 2024)

- **Surface Roughness Surrogate:** The empirical regression formula we use ($Ra = k_1 f_r^2 + k_2 VB + k_3 / v_c$) mimics their baseline objective structure (Eq. 1).

Our Novel Contribution

- **Energy-Aware Optimization:** We explicitly introduced Energy Consumption ($E = (P_{idle} + k_c \times MRR) \times t_{cut}$) as a dynamic metric in the reward function, expanding the dual-objective logic of Mo et al. into a "Green Manufacturing" policy.